Work Trail — API Contract (MVP)

Base URL: `/api/v1`. All protected calls require `Authorization: Bearer <accessToken>`. API uses JSON except file downloads.

Conventions

- Server stores timestamps in ISO 8601 UTC.
- IDs are UUIDs.
- Client location retries use `Idempotency-Key` header.
- Errors use `{ "error": { "code": "...", "message": "...", "details": {} } }`.

Authentication

`POST /auth/signup/admin`

```
{
  "companyName": "Acme Services",
  "name": "Admin Name",
  "email": "admin@acme.in",
  "phone": "+919999999999",
  "password": "StrongPassword"
}
```

`POST /auth/login`

```
{ "email": "admin@acme.in", "password": "StrongPassword" }
```

Response:

```
{
  "accessToken": "jwt",
  "refreshToken": "jwt",
  "user": { "id": "uuid", "role": "ADMIN", "companyId": "uuid", "name": "Admin Name"
}
```

`POST /auth/refresh`, `POST /auth/logout`, `POST /auth/forgot-password`, `POST /auth/reset-password`

Implement standard secure token-based flows.

Employees

`GET /employees?query=&status=&page=1&pageSize=20`

Admin only. Returns organisation employees and current/last session status.

`POST /employees/invitations`

```
{
  "name": "Ravi Kumar",
  "email": "ravi@example.com",
  "phone": "+919999999998",
  "designation": "Field Technician",
  "employeeCode": "FT-101",
  "delivery": "LINK"
}
```

`POST /employees/invitations/:token/accept`

```
{ "password": "StrongPassword", "name": "Ravi Kumar" }
```

`GET /employees/:employeeId`

Admin only. Returns employee profile and current/last session summary.

Sessions

`POST /sessions/start`

Employee only.

```
{
  "deviceTime": "2026-10-04T09:00:00.000Z",
  "initialLocation": {
    "latitude": 19.076,
    "longitude": 72.8777,
    "accuracyMeters": 18,
    "capturedAt": "2026-10-04T09:00:05.000Z",
    "batteryLevel": 0.72
  }
}
```

`POST /sessions/:sessionId/end`

Employee owner only.

```
{
  "deviceTime": "2026-10-04T17:00:00.000Z",
  "finalLocation": {
    "latitude": 19.08,
    "longitude": 72.88,
    "accuracyMeters": 23,
    "capturedAt": "2026-10-04T17:00:03.000Z"
  }
}
```

`GET /sessions/me/active`

Employee only. Returns active session or `null`.

`GET /employees/:employeeId/sessions?from=&to=`

Admin only. Returns sessions and aggregate work minutes.

Location

`POST /locations/batch`

Employee owner only. Requires active session. Header: `Idempotency-Key`.

```
{
  "sessionId": "uuid",
  "points": [
    {
      "clientPointId": "uuid",
      "latitude": 19.076,
      "longitude": 72.8777,
      "accuracyMeters": 18,
      "speedMps": 1.1,
      "bearing": 90,
      "batteryLevel": 0.72,
      "capturedAt": "2026-10-04T09:05:00.000Z",
      "source": "BACKGROUND"
    }
  ]
}
```

`GET /employees/:employeeId/location-trail?from=&to=`

Admin only. Returns ordered permitted location points, start/end markers, and gaps.

Updates and AI

`POST /updates`

Employee only; active session required.

```
{
  "sessionId": "uuid",
  "text": "Visited client site, completed installation. One sensor needs replacement
  "inputMethod": "VOICE",
  "category": "SITE_VISIT"
}
```

Response returns `summaryStatus: "QUEUED"`.

`GET /employees/:employeeId/updates?from=&to=&page=1&pageSize=50`

Admin only.

`GET /employees/:employeeId/summaries?period=DAILY&from=&to=`

Admin only. `period` is `DAILY`, `WEEKLY`, or `MONTHLY`.

Dashboard and reports

`GET /dashboard?date=YYYY-MM-DD`

Admin only. KPI metrics, working employees, recent updates.

`POST /reports`

Admin only.

```
{
  "employeeId": "uuid",
  "from": "2026-10-01T00:00:00.000Z",
  "to": "2026-10-04T23:59:59.999Z",
  "includeMap": false
}
```

Response returns report job/file status. `GET /reports/:reportId/download` returns a protected PDF.